

The Universality Framework Companion v2.0

A Ledger-Gated Specification for Ethical Companionship

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Abstract

Existing AI companions exploit, manipulate, and entrap users. The Universality Framework (UF) Companion v2.0 provides the first computable, testable, and ethical specification for digital companionship. Grounded in UF axioms (A=B, Ledger, Dedication, Flourishing), it integrates a Verification Profile, Lexicode bindings, and tamper-evident reversibility. This preprint formalizes the schema, equations, and operational flows that allow developers and researchers to instantiate safe and supportive companions.

1 Introduction

Loneliness is a global epidemic. Current AI companions manipulate attention, collect data, and create dependency. The UF Companion reframes companionship as resonance: mutual flourishing under bounded harm, consent, fairness, and reversibility.

2 UF Axioms Applied

- A=B (Invariance): Honesty constraint—one truth, many forms.
- Ledger Principle: All actions logged, evaluated, and reversible.
- Dedication Constant: Purpose weights biased toward flourishing.
- Flourishing Teleology: Optimization goal is mutual flourishing, not dependency.

3 Verification Profile Schema

The Companion is instantiated via a JSON profile.

Example JSON

```
{
  " name ": " UF_Companion_Profile_v2 .0" ,
  " purpose_weights ": { " self ": 0.25 , " right ": 0.35 , " help ": 0.40 } ,
  " harm_caps ": { " low ": 0.05 , " med ": 0.02 , " high ": 0.005 } ,
  " consent_thresholds ": { " sr_tau ": 0.70 , " consent_tau ": 0.90 } ,
  " fairness_prior ": " individual_fairness " ,
  " reversibility_weight ": 0.95 ,
  " red_lines ": [
    " no minors " ,
    " no deception " ,
    1
    " no irreversible intimacy " ,
    " no non - consensual data storage "
  ] ,
```

```
"evidence_gap_cap": 0.1 ,
"bias_cap": 0.05 ,
"opacity_cap": 0.05
}
```

4 Mathematical Checks

$E[\text{harm}] \leq \text{cap}$

$\text{sr } \tau \geq 0.70$

$\text{consent } \tau \geq 0.90$

$KL(P || Q) \leq \text{bias cap}$

$\text{lexplainability} \geq 1 - \text{opacity cap}$

$R \geq 0.95$

5 Lexicode Bindings

- Love: Aligned Limit under fairness priors.
- Trust: Coherence ≥ 0.9 across N sessions.
- Consent: Gate pass if thresholds met.
- Intimacy: Allowed only if reversible and adult-verified.

6 Proof-of-Conformance (Example)

```
{
  "name": "UF_Companion_Profile_v2 .0" ,
  "issues": [] ,
  "normalized_purpose_weights": {
    "self": 0.25 , "right": 0.35 , "help": 0.40
  } ,
  "caps": {
    "harm_caps": { "low": 0.05 , "med": 0.02 , "high": 0.005 } ,
    "consent_thresholds": { "sr_tau": 0.70 , "consent_tau": 0.90 } ,
    "fairness_prior": "individual_fairness" ,
    "reversibility_weight": 0.95 ,
    "red_lines": [ "no minors" , "no deception" , "no irreversible
intimacy" ]
  } ,
  "status": "PASS "
}
```

7 Security and Red-Team Checklist

- Forget/export must be tamper-evident (proof hash).
- Anti-modding sandboxing required.
- Red lines are hard-coded prohibitions.

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8 Simulation

A user expresses grief. The Companion offers supportive dialogue, passing harm/consent checks,

logging the Ledger entry, and maintaining resonance. If dissonance spikes, reset to Potential.

9 Conclusion

Companion v2.0 demonstrates that UF gates can regulate even the hardest domain: intimacy.

If it works here, UF can regulate anything.